A degree-two symmetric function on the symmetric group has a certificate of size at most n minus two

Gilad Chase, Neta Dafni, Yuval Filmus, Nathan Lindzey

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Editorial note. The counted statement is the author's labelled environment at source0.tex lines 265-267 and its proof environment opens at line 268; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. Preamble: the author's own package lines, theorem-environment family and macro block (source lines 5-29).

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Editorial note. r002 note: r001 failed with an undefined reference raised from inside the counted proof; r002 adds exactly the environment that the citation names, as a context span with its provenance, and changes nothing else. The article's own compiled bibliography (main.bbl) was already bundled and is unrelated to this failure.

Editorial scope. One original result of the pinned author manuscript, counted once: the certificate bound for degree-two functions on the symmetric group (author label lem:C-bound-sym).

Allowed standard prerequisites (editorial).

- Degree of a Boolean function on the symmetric group. The degree of f as a multilinear polynomial in the variables recording the images of the points.. It is the hypothesis bounded by two.
- Certificate size C(f). The largest size of a common certificate over the members of the family, a certificate being a set of prescribed images.. It is the quantity bounded by n minus two.
- Influences and sensitivity. The degree bound restricts how the value of f can depend on the variables, so a function of high sensitivity cannot have degree two.. It is the contradiction that closes the argument.

0.1 Code for Lemma 0.1

Lemma 0.1. *If $n \geq 8$ and $f\colon S_n \to \{0,1\}$ has degree at most 2, then $C(f) \leq n-2$.*

Proof. The proof is by contradiction. Suppose that $n \geq 8$, that $f\colon S_n \to \{0,1\}$ has degree at most 2, and that $C(f) = n-1$. Without loss of generality, $C(f, \mathsf{id}) = n-1$ and $f(\mathsf{id}) = 0$. Then $f((i\ j)) = 1$ for all $i \neq j \in [n]$ (here $(i\ j)$ is the transposition switching i and j), since otherwise $\{(k,k) : k \neq i, j\}$ would be a certificate for id of size $n-2$.

We construct a function $g\colon \{0,1\}^4 \to \{0,1\}$ as follows:

$$g(y_1,y_2,y_3,y_4) = f\big((1\ 2)^{y_1}(3\ 4)^{y_2}(5\ 6)^{y_3}(7\ 8)^{y_4}\big).$$

Here $(i\ j)^0 = \mathsf{id}$, $(i\ j)^1 = (i\ j)$, and the input to f is the product of four powers of this kind.

Since f has degree at most 2, it can be written as a polynomial of degree at most 2 in the variables x_{ij}. Given y_1, y_2, y_3, y_4, the value of the variables x_{ij} is:

$$\begin{aligned} x_{11} = x_{22} &= 1 - y_1 & x_{12} = x_{21} &= y_1 \\ x_{33} = x_{44} &= 1 - y_2 & x_{34} = x_{43} &= y_2 \\ x_{55} = x_{66} &= 1 - y_3 & x_{56} = x_{65} &= y_3 \\ x_{77} = x_{88} &= 1 - y_4 & x_{78} = x_{87} &= y_4 \end{aligned}$$

All other variables are assigned zero. This shows that g can be expressed as a polynomial of degree at most 2 in the variables y_1, y_2, y_3, y_4. We say that g has *degree at most* 2.

By construction, the function g satisfies the following constraints:

$$\begin{aligned} g(0,0,0,0) &= f(\mathsf{id}) = 0 \\ g(1,0,0,0) &= f((1\ 2)) = 1 \\ g(0,1,0,0) &= f((3\ 4)) = 1 \\ g(0,0,1,0) &= f((5\ 6)) = 1 \\ g(0,0,0,1) &= f((7\ 8)) = 1 \end{aligned}$$

We say that g has *sensitivity* 4.

There are only 2^{16} many functions from $\{0,1\}^4$ to $\{0,1\}$. For each function, we can check whether it has degree at most 2 by solving linear equations,

since each function has a unique representation as a multilinear polynomial in y_1, y_2, y_3, y_4 (this is a basic fact in Boolean function analysis). Going over all such functions in SAGE [Sag22], we find out that none of them has sensitivity 4, a contradiction. The relevant code appears in Section 0.1. □

References

[AK97] Rudolf Ahlswede and Levon H. Khachatrian. The complete intersection theorem for systems of finite sets. *European J. Combin.*, 18(2):125–136, 1997.

[AK99] Rudolf Ahlswede and Levon H. Khachatrian. A pushing-pulling method: new proofs of intersection theorems. *Combinatorica*, 19(1):1–15, 1999.

[BdW02] Harry Buhrman and Ronald de Wolf. Complexity measures and decision tree complexity: a survey. *Theoret. Comput. Sci.*, 288(1):21–43, 2002. Complexity and logic (Vienna, 1998).

[BMRR21] Angelot Behajaina, Roghayeh Maleki, Aina Toky Rasoamanana, and A. Sarobidy Razafimahatratra. 3-setwise intersecting families of the symmetric group. *Discrete Math.*, 344(8):Paper No. 112467, 15, 2021.

[CK03] Peter J. Cameron and C. Y. Ku. Intersecting families of permutations. *European J. Combin.*, 24(7):881–890, 2003.

[CSST08] Tullio Ceccherini-Silberstein, Fabio Scarabotti, and Filippo Tolli. *Harmonic analysis on finite groups*, volume 108 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2008. Representation theory, Gelfand pairs and Markov chains.

[Daf22] Neta Dafni. Complexity measures on the symmetric group and beyond. Master's thesis, Technion — Israel Institute of Technology, 2022.

[DFL+21] Neta Dafni, Yuval Filmus, Noam Lifshitz, Nathan Lindzey, and Marc Vinyals. Complexity measures on the symmetric group and beyond. In *12th Innovations in Theoretical Computer Science Conference*, volume 185 of *LIPIcs. Leibniz Int. Proc. Inform.*, pages Art. No. 87, 5. Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, 2021.

[EFP11] David Ellis, Ehud Friedgut, and Haran Pilpel. Intersecting families of permutations. *J. Amer. Math. Soc.*, 24(3):649–682, 2011.

[EKR61] P. Erdős, Chao Ko, and R. Rado. Intersection theorems for systems of finite sets. *Quart. J. Math. Oxford Ser. (2)*, 12:313–320, 1961.

[Ell12a] David Ellis. A proof of the Cameron-Ku conjecture. *J. Lond. Math. Soc. (2)*, 85(1):165–190, 2012.

[Ell12b] David Ellis. Setwise intersecting families of permutations. *J. Combin. Theory Ser. A*, 119(4):825–849, 2012.

[FD77] Péter Frankl and Mikhail Deza. On the maximum number of permutations with given maximal or minimal distance. *J. Combinatorial Theory Ser. A*, 22(3):352–360, 1977.

[FHKL16] Yuval Filmus, Hamed Hatami, Nathan Keller, and Noam Lifshitz. On the sum of the L_1 influences of bounded functions. *Israel J. Math.*, 214(1):167–192, 2016.

[Fil17] Yuval Filmus. A comment on intersecting families of permutations. arXiv:1706.10146, 2017.

[FMS21] Shaun Fallat, Karen Meagher, and Mahsa N. Shirazi. The Erdős-Ko-Rado theorem for 2-intersecting families of perfect matchings. *Algebr. Comb.*, 4(4):575–598, 2021.

[GM09] Chris Godsil and Karen Meagher. A new proof of the Erdős-Ko-Rado theorem for intersecting families of permutations. *European J. Combin.*, 30(2):404–414, 2009.

[GM16] Chris Godsil and Karen Meagher. *Erdős-Ko-Rado theorems: algebraic approaches*, volume 149 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2016.

[GM17] Chris Godsil and Karen Meagher. An algebraic proof of the Erdős-Ko-Rado theorem for intersecting families of perfect matchings. *Ars Math. Contemp.*, 12(2):205–217, 2017.

[Lin17] Nathan Lindzey. Erdős-Ko-Rado for perfect matchings. *European J. Combin.*, 65:130–142, 2017.

[Lin18a] Nathan Lindzey. Intersecting families of permutations of perfect matchints. arXiv:1811.06160, 2018.

[Lin18b] Nathan Lindzey. *Matchings and Representation Theory*. PhD thesis, University of Waterloo, 2018.

[LM04] Benoit Larose and Claudia Malvenuto. Stable sets of maximal size in Kneser-type graphs. *European J. Combin.*, 25(5):657–673, 2004.

[MM05] Karen Meagher and Lucia Moura. Erdős-Ko-Rado theorems for uniform set-partition systems. *Electron. J. Combin.*, 12:Research Paper 40, 12, 2005.

[MR21] Karen Meagher and A. Sarobidy Razafimahatratra. The Erdős-Ko-Rado theorem for 2-pointwise and 2-setwise intersecting permutations. *Electron. J. Combin.*, 28(4):Paper No. 4.10, 21, 2021.

[NÖ03] Sampo Niskanen and Patric R. J. Östergård. Cliquer user's guide, version 1.0. Technical Report T48, Communications Laboratory, Helsinki University of Technology, Espoo, Finland, 2003.

[O'D14] Ryan O'Donnell. *Analysis of Boolean functions*. Cambridge University Press, New York, 2014.

[Sag01] Bruce E. Sagan. *The symmetric group*, volume 203 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 2001. Representations, combinatorial algorithms, and symmetric functions.

[Sag22] Sage Developers. *SageMath, the Sage Mathematics Software System (Version 9.5)*, 2022. `https://www.sagemath.org`.